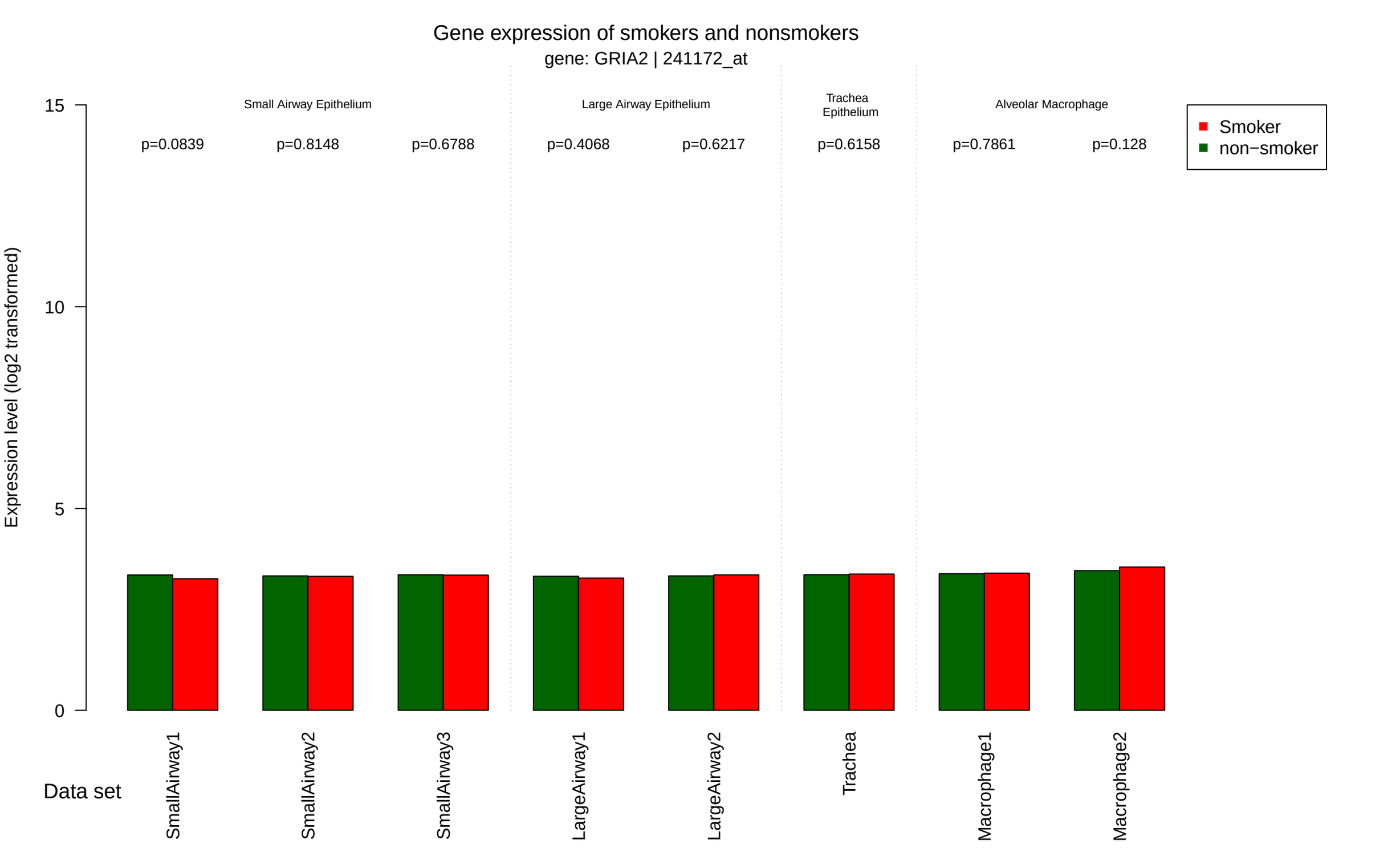
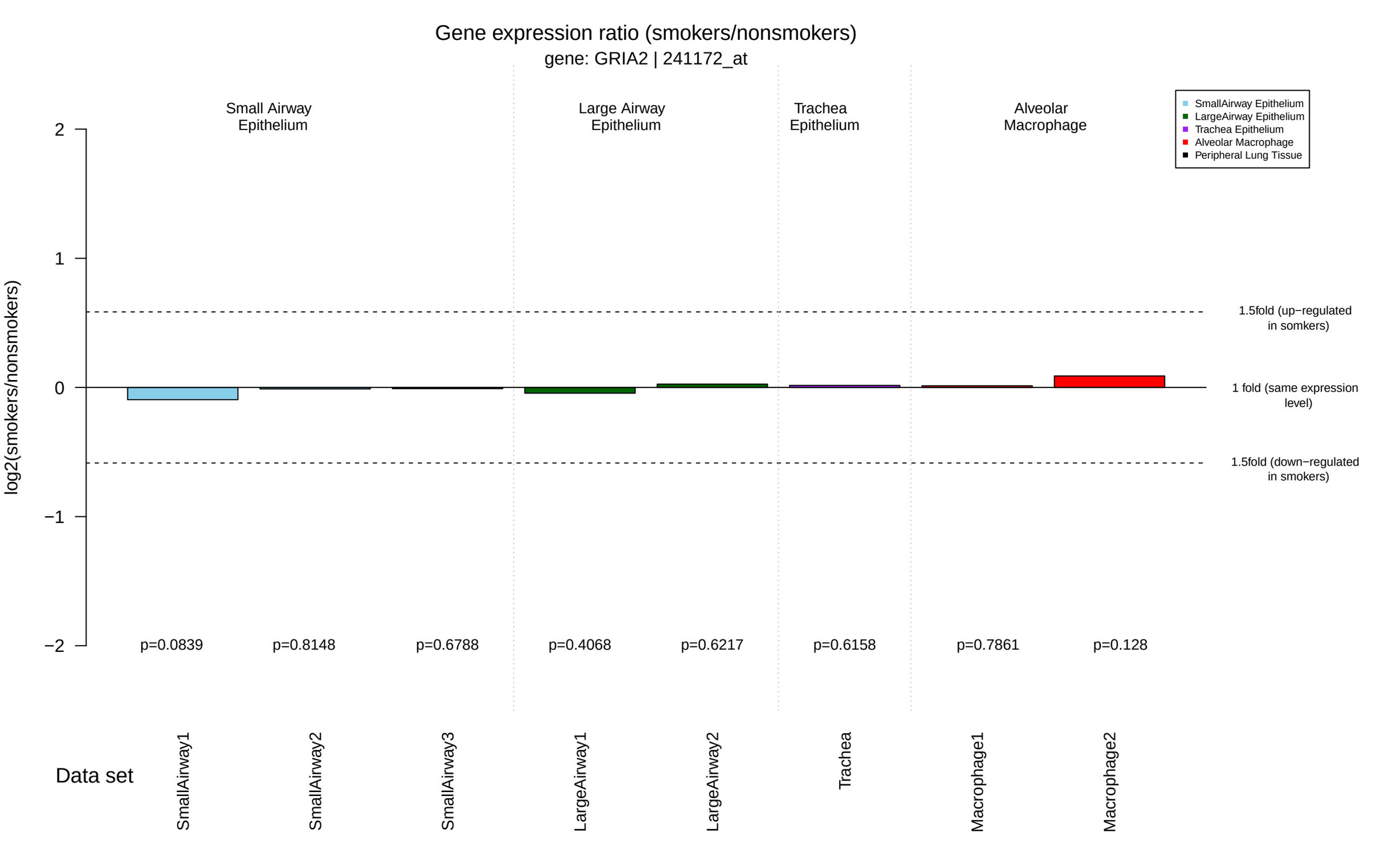
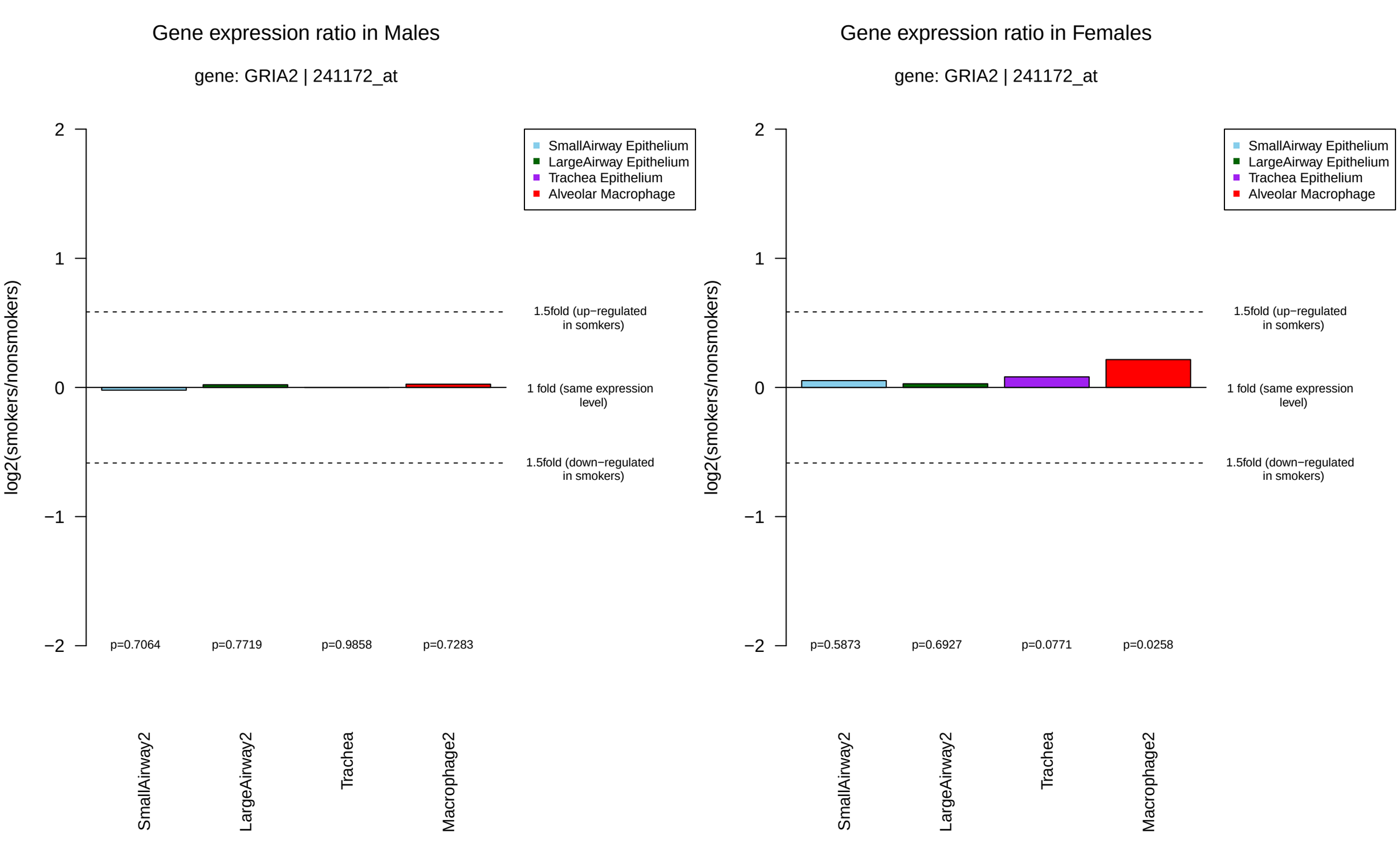
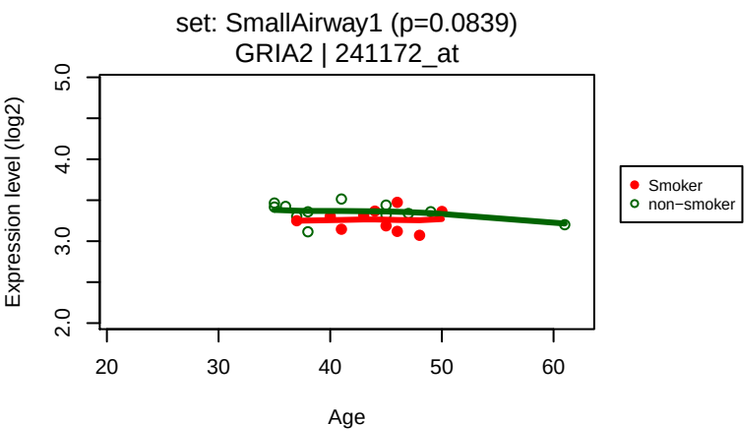
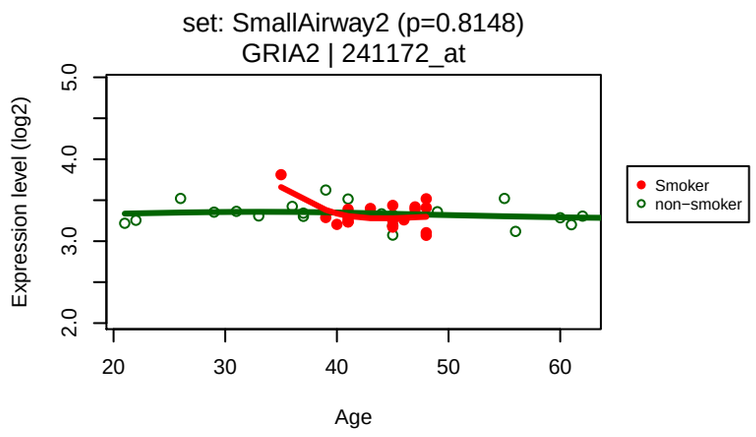
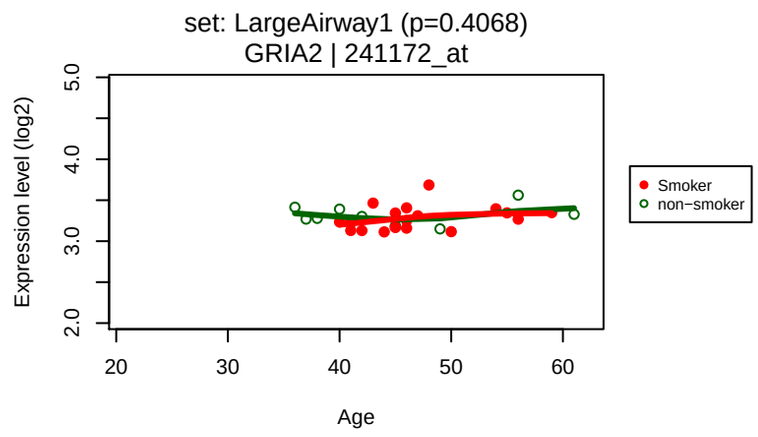

Gene expression of smokers and nonsmokers
gene: GRIA2 | 241172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0839
p=0.8148
p=0.6788
p=0.4068
p=0.6217
p=0.6158
p=0.7861
p=0.128
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

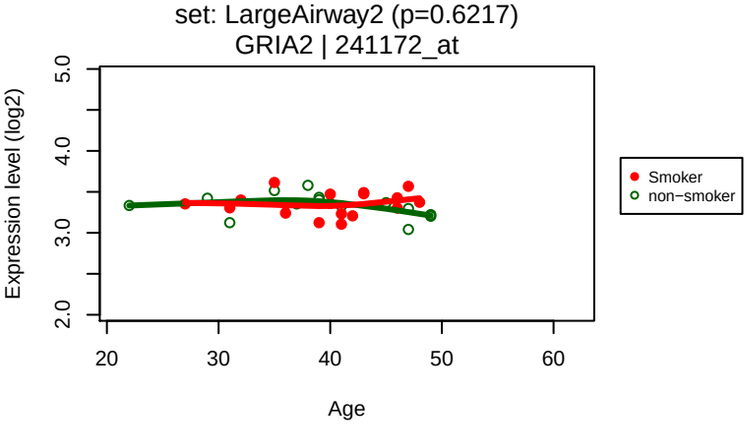

Gene expression ratio (smokers/nonsmokers)
gene: GRIA2 | 241172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0839
p=0.8148
p=0.6788
p=0.4068
p=0.6217
p=0.6158
p=0.7861
p=0.128
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

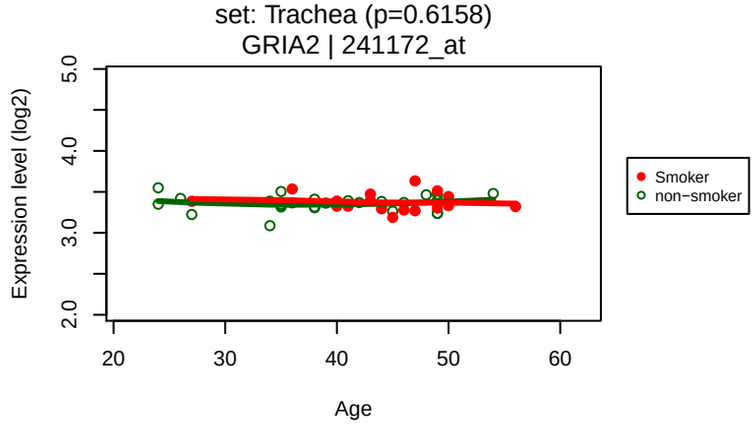

Gene expression ratio in Males
gene: GRIA2 | 241172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7064
p=0.7719
p=0.9858
p=0.7283
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GRIA2 | 241172_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5873
p=0.6927
p=0.0771
p=0.0258
SmallAirway2
LargeAirway2
Trachea
Macrophage2

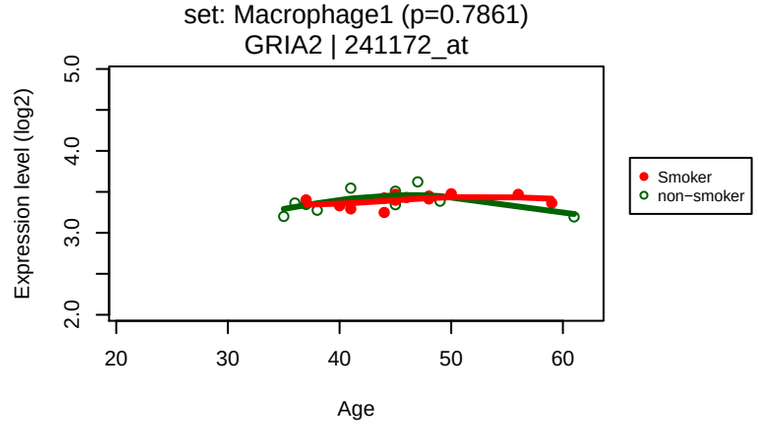

set: SmallAirway1 (p=0.0839)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker

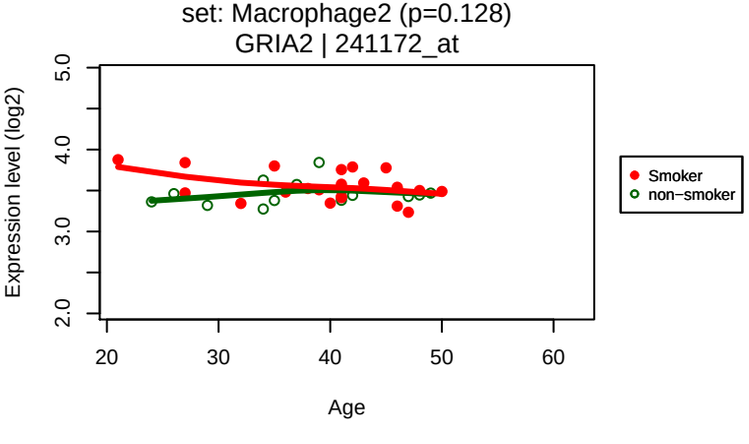

set: SmallAirway2 (p=0.8148)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker

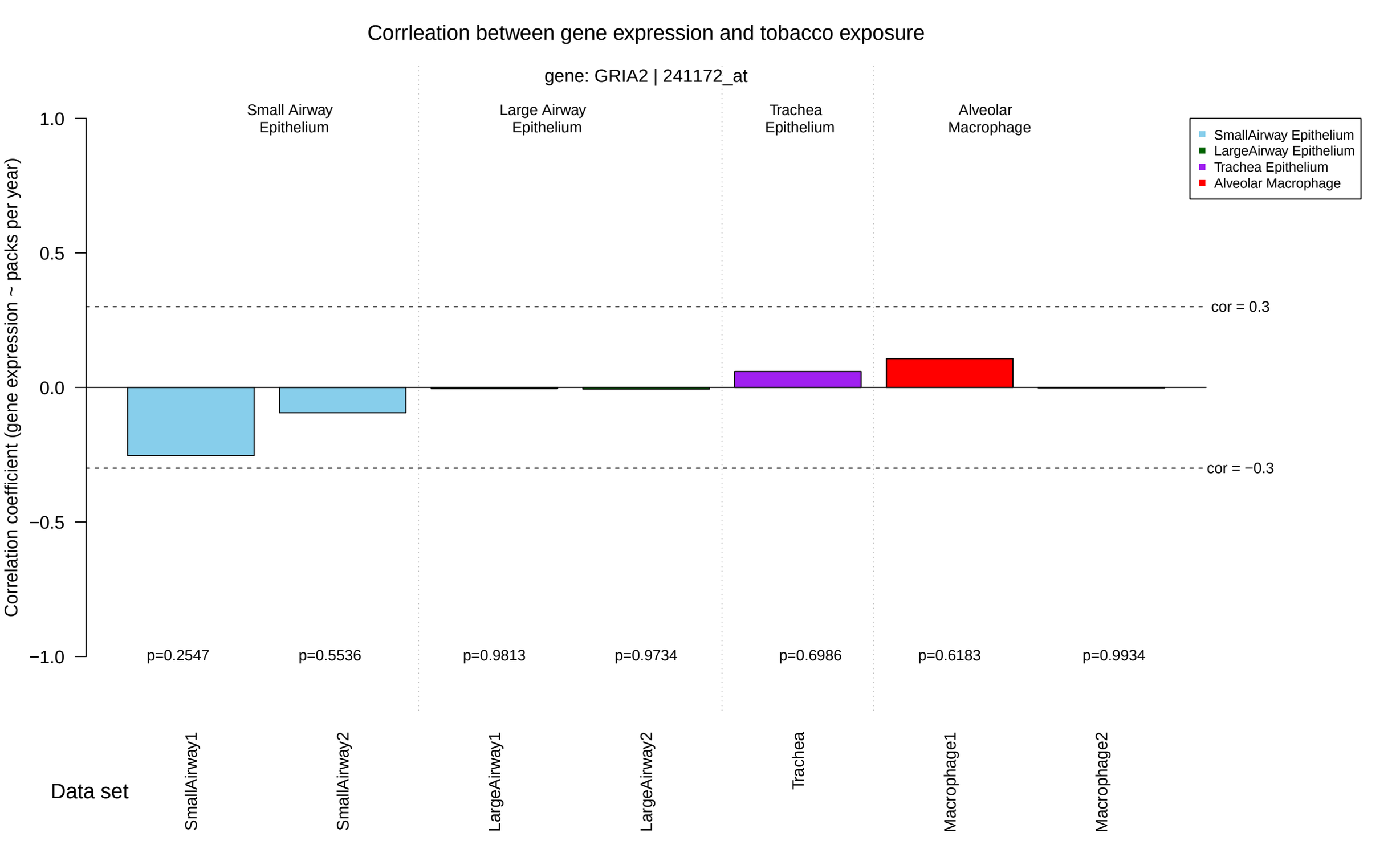

set: LargeAirway1 (p=0.4068)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6217)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6158)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7861)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.128)
GRIA2 | 241172_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GRIA2 | 241172_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2547
p=0.5536
p=0.9813
p=0.9734
p=0.6986
p=0.6183
p=0.9934
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2